



Acceptable Encryption Standard

(Last Updated February 2026)

Purpose

The purpose of this policy is to provide guidance that limits the use of encryption to those algorithms that have received substantial public review and have been proven to work effectively. Additionally, this policy provides direction to ensure that Federal regulations are followed, and legal authority is granted for the dissemination and use of encryption technologies outside of the United States.

Scope

This policy applies to all **Phour Global Limited** employees and affiliates.

Safeguards

Algorithm Requirements

Ciphers in use must meet or exceed the set defined as “AES-compatible” or “partially AES-compatible” according to the IETF/IRTF Cipher Catalog, or the set defined for use in the United States National Institute of Standards and Technology (NIST) publication FIPS 140-2, or any superseding documents according to the date of implementation. The use of the Advanced Encryption Standard (AES) is strongly recommended for symmetric encryption.



Algorithms in use must meet the standards defined for use in NIST publication FIPS 140-2 or any superseding document, according to the date of implementation. The use of the RSA and Elliptic Curve Cryptography (ECC) algorithms is strongly recommended for asymmetric encryption.

Signature Algorithms

Algorithm	Key Length (min)	Additional Comment
ECDSA	P-256	Consider RFC6090 to avoid patent infringement.
RSA	2048	Must use a secure padding scheme. PKCS#7 padding scheme is recommended. Message hashing required.
LDWM	SHA256	Refer to LDWM Hash-based Signatures Draft

Hash Function Requirements

In general, **Phour Global Limited** adheres to the NIST Policy on **Hash Functions**

Key Agreement and Authentication

Key exchanges must use one of the following cryptographic protocols: Diffie-Hellman, IKE, or Elliptic Curve Diffie-Hellman (ECDH).

End points must be authenticated prior to the exchange or derivation of session keys.



Public keys used to establish trust must be authenticated prior to use. Examples of authentication include transmission via cryptographically signed message or manual verification of the public key hash.

All servers used for authentication (for example, RADIUS or TACACS) must have installed a valid certificate signed by a known trusted provider.

All servers and applications using SSL or TLS must have the certificates signed by a known, trusted provider.

Key Generation

Cryptographic keys must be generated and stored in a secure manner that prevents loss, theft, or compromise.

Key generation must be seeded from an industry standard random number generator (RNG). For examples, see [NIST Annex C: Approved Random Number Generators for FIPS PUB 140-2](#)

Policy Sanctions

Non-compliance with this policy may result in disciplinary action in line with the human resources procedures of **Phour Global Limited**. Consequences may range from mandatory refresher training and written warnings to temporary suspension of remote access privileges and, in severe cases, termination of employment or contractual obligations. Individuals may be subject to legal consequences under applicable laws if violations involve illegal activities. These sanctions emphasize the critical importance of cybersecurity, the individual's role in protecting the digital assets of **Phour Global Limited**, and the potential risks associated with policy



violations. Enforcement will be consistent and impartial, with the severity of the action corresponding directly to the seriousness of the breach.

